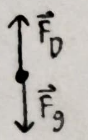
6-17-25

PHY201 Exam on chapter(s): 4, 5, 6

Name:

These questions assess your understanding of the material from Chapters 4, 5, 6 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. **Stephen, who has a mass of 95 kg, has jumped from a plane and is falling through the air at his terminal velocity, a speed of 57 m/s. The temperature of the air is 11°C. Recall that the density of air ρ (in kg/m³) depends on temperature T (in °C) according to $\rho = \frac{353}{T+273}$. As Stephen moves over a distance of 3.8 m through the air, calculate his area that faces the air. Assuming a drag coefficient of 0.96.**



$\Sigma F_y = F_D - F_g = m\cancel{a_y}$

$F_D = F_g = mg$

$F_D = \frac{1}{2}C\rho A v^2$

$mg = \frac{1}{2}C\rho A v^2$

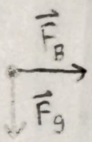
$A = \frac{2mg}{C\rho v^2}$

$A = \frac{2(95\ kg)(9.8\ \frac{m}{s^2})}{(0.96)\left(\frac{353}{11+273}\ \frac{kg}{m^3}\right)\left(57\ \frac{m}{s}\right)^2} = \boxed{0.48\ m^2}$

$\frac{kg\cdot m}{s^2}\cdot\frac{m^3\cdot s^2}{kg\cdot m^2} = m^2$

ANSWER: 0.48 m²

2. **A flea jumps into the air by exerting a force straight down on the ground. A breeze blows on the flying flea and exerts a horizontal force of 6 μN on the flea. Find the magnitude of the acceleration of the flea if its mass is 3.8 mg.**



$\Sigma F_x = F_B = ma_x$

$\Sigma F_y = -F_g = -mg = ma_y$

$F = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$

$F = ma$

$a = \frac{\sqrt{(F_B)^2 + (-mg)^2}}{m}$

$a = \frac{\sqrt{(6\times10^{-6}\ N)^2 + ((-3.8\times10^{-6}\ kg)(9.8\ \frac{m}{s^2}))^2}}{3.8\times10^{-6}\ kg} = \boxed{9.93\ \frac{m}{s^2}}$

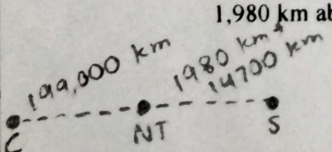
$\frac{6\ \mu N}{} \Big| \frac{1\ N}{10^6\ \mu N} = 6\times10^{-6}\ N$

$\frac{3.8\ mg}{} \Big| \frac{1\ kg}{10^6\ mg} = 3.8\times10^{-6}\ kg$

$\sqrt{\frac{kg^2\cdot m^2}{s^4} + \frac{kg^2\cdot m^2}{s^4}} = \frac{kg\cdot m}{s^2}$

ANSWER: 9.93 m/s²

3. **Given that the planet Celestia orbits the planet Nova Terra once every 110.4 days and that it is an average distance of 199,000 km from the center of Nova Terra, calculate the period of an artificial satellite orbiting Nova Terra at an average altitude of 1,980 km above Nova Terra's surface. Note that the radius of Nova Terra is 14,700 km.**



$\frac{T_s^2}{T_c^2} = \frac{r_s^3}{r_c^3}$

$T_s = \sqrt{\frac{r_s^3\cdot T_c^2}{r_c^3}}$

$T_s = \sqrt{\frac{(1980\ km + 14700\ km)^3(110.4\ days)^2}{(199{,}000\ km)^3}} = \boxed{2.68\ days}$

ANSWER: 2.68 days

+3

4. **As Grace waits for her food to heat up, she determines that her microwave oven rotates the food with a speed of 12.8 revolutions/minute. What is the food's angular velocity in radians per second?**

ANSWER: $1.34\ \frac{rad}{s}$

$$\frac{12.8\ \cancel{rev}}{1\ \cancel{min}} \cdot \frac{2\pi\ rad}{1\ \cancel{rev}} \cdot \frac{1\ \cancel{min}}{60\ s} = \boxed{1.34\ \frac{rad}{s}}$$

5. **Alexis is not only a part-time guide at the Battleship USS Iowa Museum, but also a physics student. She wants to find the net external force that would be exerted on a 1210-kg artillery shell fired from one of the USS Iowa's turrets. Alexis measures the barrel of a turret to be 17.0-m long and reads that a shell, fired from rest, would leave the barrel at 760 m/s. What is the net external force exerted on the shell?**

ANSWER: 2.06×10^7 N

$\Sigma F_x = ma_x$

$\vec{v}_{xf}^{\,2} = \vec{v}_{xi}^{\,2} + 2\vec{a}_x \Delta\vec{x}$

$\vec{a}_x = \frac{\vec{v}_{xf}^{\,2}}{2\Delta\vec{x}}$

$\Sigma F_x = \frac{m\vec{v}_{xf}^{\,2}}{2\Delta\vec{x}}$

$\Sigma F_x = \frac{(1210\ kg)(760\ \frac{m}{s})^2}{2(17\ m)} = \boxed{2.06 \times 10^7\ N}$

6. **A 17-kg child is riding a playground merry-go-round that is rotating at 6.0 revolutions per minute. What centripetal force must she exert to stay on if she is 6.3 m from the center of the merry-go-round?**

ANSWER: 42.28 N

$$\frac{6\ \cancel{rev}}{1\ \cancel{min}} \cdot \frac{2\pi\ rad}{1\ \cancel{rev}} \cdot \frac{1\ \cancel{min}}{60\ s} = 0.628\ \frac{rad}{s}$$

$\vec{F}_c = m\vec{a}_c$

$a_c = r\omega^2$

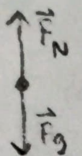
$F_c = mr\omega^2$

$F_c = (17\ kg)(6.3\ m)(0.628\ \frac{rad}{s})^2 = \boxed{42.28\ N}$

+3

7. Victoria, who has a mass of 77.4 kg, steps onto a scale that rests on the floor of an elevator. If the elevator accelerates upward at a rate of 6.6 m/s², what will the scale read?

ANSWER: 1269.36 N ✓



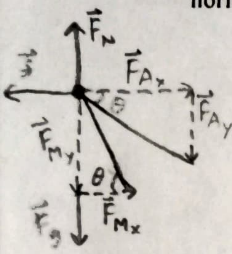
$\Sigma F_y = F_N - F_g = ma_y$

$F_N = ma_y + mg$

$F_N = (77.4\ \text{kg})(6.6\ \frac{m}{s^2}) + (77.4\ \text{kg})(9.8\ \frac{m}{s^2}) = \boxed{1269.36\ N}$

8. Two nurses, Mary and Allison, work together to push a crash cart by pushing on the handle. They push it from rest so that it covers a distance of 5.5 m in 6.0 s. Mary exerts 190 N at 46.7° below the horizontal and Allison exerts 100 N at 25.1° below the horizontal. The force of friction is 93 N. Determine the mass of the cart.

ANSWER: 418.46 kg ✓



$\Sigma F_x = F_{A_x} + F_{M_x} - f = ma_x$

$F_A \cos_A + F_M \cos_M - f = ma_x$

$\vec{x}_f = \vec{x}_i + \vec{v}_{x_i} t + \frac{1}{2}\vec{a}_x t^2$

$a_x = \frac{2\vec{x}_f}{t^2}$

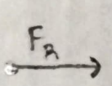
$m = \frac{F_A \cos_A + F_M \cos_M - f}{\frac{2\vec{x}_f}{t^2}}$

$m = \frac{(100\ N)\cos(25.1°) + (190\ N)\cos(46.7°) - 93\ N}{\frac{2(5.5\ m)}{(6\ s)^2}}$

$\boxed{m = 418.46\ \text{kg}}$

9. Robert and Shiv are playing rugby. Robert pushes Shiv with a force of 1326 N. Robert has a mass of 60 kg while Shiv has a mass of 63 kg. What is Shiv's acceleration as a result of getting hit by Robert?

ANSWER: 21.05 $\frac{m}{s^2}$ ✓



$\Sigma F_x = m_s a_x$

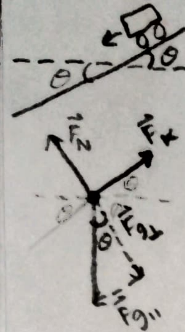
$a_x = \frac{\Sigma F_x}{m_s}$

$a_x = \frac{1326\ N}{63\ \text{kg}} = \boxed{21.05\ \frac{m}{s^2}}$

+3

10. Nicholas is trying to keep a 133.9-kg wagon at rest on a frictionless inclined plane, but despite his best efforts, he can only apply a force of 575 N to try to stop the motion. If the plane makes an angle of 42.4° with the horizontal, what is the acceleration of the wagon down the plane?

ANSWER: $2.31\ \frac{m}{s^2}$



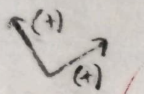
$F_k = F_{Nicholas}$

$\Sigma F_{\parallel} = F_k - F_{g\parallel} = m(-a_{\parallel})$

$F_{g\parallel} = F_g \sin\theta = mg\sin\theta$

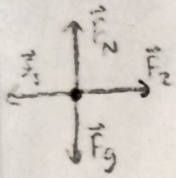
$a_{\parallel} = \frac{F_k - mg\sin\theta}{-m}$

$a_{\parallel} = \frac{(575\ N) - (133.9\ kg)(9.8\ \frac{m}{s^2})\sin(42.4°)}{-(133.9\ kg)} = \boxed{2.31\ \frac{m}{s^2}}$



11. Zoe pushes a crate made of stellanium on a floor made of aromatic cedar. The crate is initially at rest on the horizontal floor, until Zoe pushes horizontally on the crate with a force of 320 N, the minimum force necessary to start its motion. Find the mass of the crate. Note that for stellanium on aromatic cedar, $\mu_s = 0.55$ and $\mu_k = 0.52$.

ANSWER: 59.37 kg



$\Sigma F_y = F_N - F_g = ma_y$

$F_N = F_g = mg$

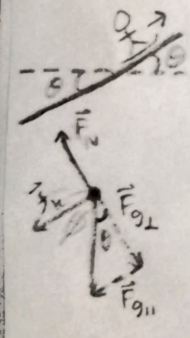
$\Sigma F_x = F_2 - f_{s,max} = ma_x$

$F_2 = f_{s,max} = \mu_s F_N = \mu_s mg$

$m = \frac{F_2}{\mu_s g} = \frac{320\ N}{(0.55)(9.8\ \frac{m}{s^2})} = \boxed{59.37\ kg}$

12. Anna is decelerating as she snowboards at 4.0 m/s up a slope that is inclined at 15.7° above the horizontal. Calculate the magnitude of her deceleration. The total mass of Anna and her snowboard is 93 kg. The density of the air is 1.289 kg/m³. Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$. Ignore the effect of air resistance.

ANSWER: $3.595\ \frac{m}{s^2}$



$\Sigma F_{\perp} = F_N - F_{g\perp} = ma_{\perp}$

$F_N = F_{g\perp} = F_g\cos\theta = mg\cos\theta$

$\Sigma F_{\parallel} = -f_k - F_{g\parallel} = m(-a_{\parallel})$

$-\mu_k F_N - mg\sin\theta = -ma_{\parallel}$

$a_{\parallel} = \frac{-\mu_k mg\cos\theta - mg\sin\theta}{-m}$

$a_{\parallel} = \frac{-(0.1)(9.8\ \frac{m}{s^2})\cos(15.7°) - (9.8\ \frac{m}{s^2})\sin(15.7°)}{-1} = \boxed{3.595\ \frac{m}{s^2}}$